

THE MOUNTAIN PATH

World of Finance

PRINCIPAL COMPONENT ANALYSIS (PCA)

A Complete Learning Guide

Every Step Explained – From Intuition to Calculation

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28+ Years Corporate Finance & Banking Experience

10+ Years Academic Excellence

Educational Series

For MBA, CFA & FRM Students

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Contents

Introduction: Why PCA Matters	3
1 Multiple Linear Regression: A Quick Recap	7
1.1 What Is Multiple Linear Regression (MLR)?	7
1.2 The MLR Equation — Explained Piece by Piece	7
1.3 How Are the Coefficients (β) Estimated?	8
1.4 The Five Key Assumptions of MLR	9
1.5 Model Evaluation Metrics	9
2 The Problem with MLR: Multicollinearity	10
2.1 What Is Multicollinearity?	10
2.2 Why Is Multicollinearity a Problem? — Every Consequence Explained	11
2.3 How Do We Detect Multicollinearity?	11
2.3.1 Method 1: Correlation Matrix	11
2.3.2 Method 2: Variance Inflation Factor (VIF)	12
2.3.3 Method 3: Eigenvalues Near Zero	12
2.3.4 Method 4: Condition Number	12
2.4 The Usual Fixes (and Why PCA Is Special)	13
3 PCA — The Intuition (Before Any Math)	14
3.1 The Photography Analogy	14
3.2 The Diagonal Cloud Analogy	14
3.3 What Is a Principal Component?	15
3.4 The Three Magic Properties of Principal Components	15
4 The Math Behind PCA — A Deep Dive	17
4.1 Step 1: The Mean — Foundation of Everything	17
4.2 Step 2: Variance — Measuring Spread	17
4.3 Step 3: Covariance — How Two Variables Move Together	18
4.4 Step 4: The Covariance Matrix Σ	18
4.5 Step 5: Eigenvalues and Eigenvectors — The Core of PCA	20
4.5.1 What Is an Eigenvector? What Is an Eigenvalue?	20
4.5.2 Why Do Eigenvalues Matter for PCA?	20
4.5.3 How to Calculate Eigenvalues — The General Method	21
4.5.4 The Closed-Form Solution for a 2×2 Matrix	21
4.5.5 Complete Numerical Calculation with Our Data	23
4.5.6 How to Calculate Eigenvectors	24
4.6 Step 6: Explained Variance — How Important Is Each PC?	25
4.7 Step 7: Computing PC Scores	25
5 PCA Step-by-Step Recipe	27
5.1 Choosing the Number of Components (k) — Four Methods Explained	27
6 Principal Component Regression (PCR)	29
6.1 The PCR Equation — Explained	29
6.2 Why Does PCR Cure Multicollinearity?	29
6.3 Translating Back to Original Variables	29

6.4	PCR Workflow Cheat Sheet	30
6.5	PCR vs. Ridge vs. PLS — When to Use What	30
7	Worked Illustration: Complete 2-Variable PCA	31
7.1	The Complete Data Table	31
7.2	Summary of All Calculations	31
7.3	Visual: Before vs. After PCA	32
8	Pros, Cons, and Pitfalls of PCR	33
8.1	Five Common Pitfalls — Explained in Detail	33
9	Glossary and FAQ	35
9.1	Glossary of Key Terms	35
9.2	Frequently Asked Questions	35
10	Case Study: Predicting House Prices with PCR	37
10.1	Step 1: The Raw Data ($n = 12$ houses, $p = 5$ features)	37
10.2	Step 2: Detect the Problem — The Correlation Matrix	37
10.3	Step 3: Standardise All Predictors	38
10.4	Step 4: Eigen-Decomposition Results	38
10.5	Step 5: Loadings — Interpreting What Each PC Means	39
10.6	Step 6: PCR Results — Regression on PC Scores	39
10.7	Step 7: The Verdict — MLR vs. PCR Head-to-Head	40
11	Decision Framework: When Should You Use PCA?	41
12	Abbreviations and Acronyms	42
13	About the Author	43

Introduction: Why PCA Is One of the Most Important Techniques in Modern Analytics

The Data Explosion Problem

We live in an age of high-dimensional data. A credit risk model may use 200+ financial ratios. A machine learning image classifier processes millions of pixel values. A macroeconomic forecasting model draws on hundreds of correlated indicators — GDP, inflation, employment, interest rates, trade balances, commodity prices, and more.

The challenge is clear: **more variables do not automatically mean better models**. In fact, adding correlated or noisy variables often makes models *worse* — a phenomenon known as the **curse of dimensionality**. Coefficients become unstable, computation slows, overfitting increases, and interpretability collapses.

Principal Component Analysis (PCA) is the single most widely used solution to this problem. Developed by Karl Pearson in 1901 and formalised by Harold Hotelling in 1933, PCA has become a foundational tool across statistics, finance, engineering, and machine learning.

PCA in Multiple Linear Regression (MLR)

The MLR Problem That PCA Solves

In Multiple Linear Regression, the OLS estimator is:

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

When predictor columns of $\mathbf{X}$ are highly correlated, $(\mathbf{X}^\top \mathbf{X})$ becomes **nearly singular**. Its inverse amplifies small data perturbations into massive coefficient swings. Standard errors explode, p -values become meaningless, and the model — though fitting well on R^2 — is fundamentally unreliable for inference.

PCA transforms correlated predictors into a smaller set of uncorrelated principal components, making $(\mathbf{Z}^\top \mathbf{Z})$ perfectly diagonal and trivially invertible. The result: **stable coefficients, honest standard errors, and a model you can trust**.

This is not merely an academic exercise. In practice, multicollinearity is the norm, not the exception:

- In **credit scoring**, income, assets, debt-to-income, and credit utilisation are all intercorrelated.
- In **equity factor models**, size, value, momentum, and quality factors share common drivers.
- In **macroeconomic forecasting**, GDP growth, industrial production, consumer spending, and employment are tightly linked.

Without PCA (or a similar technique), regression on these datasets produces models that look good on paper but fail in deployment.

PCA in Machine Learning

PCA's role extends far beyond classical regression. In modern machine learning, it serves as a critical **preprocessing** and **feature engineering** step:

PCA's Five Roles in Machine Learning

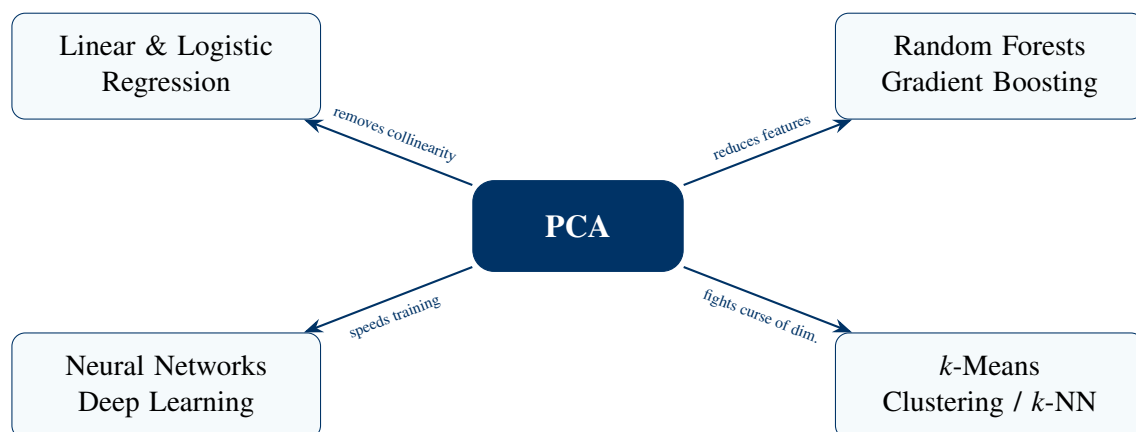
1. Dimensionality Reduction (Feature Compression): High-dimensional datasets (hundreds or thousands of features) are computationally expensive and prone to overfitting. PCA compresses them into a compact representation. For example, reducing 500 financial features to 20 principal components that capture 95% of the variance dramatically speeds up training while preserving predictive power.

2. Noise Removal (Signal Extraction): The last few principal components typically capture noise rather than signal. By discarding them, PCA acts as a built-in denoising filter. This improves generalisation — the model learns the true underlying pattern rather than memorising noise.

3. Multicollinearity Elimination: ML algorithms like logistic regression, linear SVMs, and neural networks with linear layers all suffer when features are correlated. PCA guarantees orthogonal (uncorrelated) inputs, stabilising gradient-based optimisation and coefficient estimation.

4. Visualisation of High-Dimensional Data: Reducing data to 2 or 3 PCs allows scatter-plot visualisation of clusters, outliers, and class separability in datasets with dozens of original features. This is invaluable for exploratory data analysis before model building.

5. Preventing the Curse of Dimensionality: As dimensions grow, data points become increasingly sparse. Distance metrics lose meaning, and algorithms like k -NN and clustering degrade. PCA projects data into a lower-dimensional subspace where these algorithms regain effectiveness.



PCA in Forecasting Models

In time-series forecasting and financial modelling, PCA plays a distinctive and powerful role:

Illustration

Yield Curve Modelling: The term structure of interest rates (2Y, 5Y, 10Y, 30Y yields, etc.) involves many highly correlated maturities. PCA typically reveals that **three components** — Level (parallel shifts), Slope (steepening/flattening), and Curvature (butterfly) — explain over 99% of yield curve movements. Central banks, fixed-income desks, and risk managers worldwide use this PCA decomposition daily.

Factor Models in Equity Markets: PCA on stock return covariance matrices identifies latent factors driving the cross-section of returns, forming the basis of statistical arbitrage strategies and risk models (e.g., BARRA, Axioma).

Macroeconomic Nowcasting: When forecasting GDP or inflation, economists face hundreds of candidate indicators. PCA extracts a handful of “diffusion indices” that summarise the state of the economy, as pioneered by Stock and Watson (2002). The Federal Reserve’s factor-augmented VAR (FAVAR) models rely on exactly this approach.

Value-at-Risk (VaR): Portfolio risk models must handle the covariance of hundreds of positions. PCA reduces this to a manageable set of risk factors, making Monte Carlo simulation and scenario analysis computationally feasible.

Credit Risk Modelling: In credit scoring with 100+ borrower attributes, PCA compresses correlated features (income, debt ratios, payment history metrics) into orthogonal components, improving logistic regression stability and scorecard performance.

When to Use PCA — and When Not To**Use PCA when:**

- You have many correlated variables and need dimensionality reduction.
- You want to visualise high-dimensional data.
- You want to reduce noise before modelling.
- Compact representation matters more than interpreting individual original variables.
- You face the curse of dimensionality (p large relative to n).

Do NOT use PCA when:

- Variables are already uncorrelated and few in number — plain regression suffices.
- Interpreting individual feature effects is critical (consider Ridge or LASSO instead).
- Data contains primarily categorical variables (use MCA instead).
- Non-linear relationships dominate (consider kernel PCA or autoencoders).

What This Guide Covers

This guide takes you on a complete journey through PCA:

Section	Topic	What You Will Learn
1	MLR Recap	The regression framework PCA supports
2	The Problem	Why multicollinearity breaks MLR
3	PCA Intuition	What PCA does, without any math
4	The Math	Covariance, eigenvalues, eigenvectors — from scratch
5	Step-by-Step Recipe	The 7-step PCA workflow
6	PCR	Combining PCA with MLR
7	Worked Illustration	Full 2-variable numerical example
8	Pros, Cons & Pitfalls	When PCR shines and when it hurts
9	Glossary & FAQ	Quick reference and exam-style questions
10	Case Study	End-to-end house price prediction with PCR

Every concept is explained from first principles with worked calculations, “why” boxes, and visual illustrations. Whether you are an MBA student, a CFA/FRM candidate, or a data science practitioner, this guide is designed to make PCA not just *understood* but *intuitive*.

*“The goal is not just to reduce dimensions
— it is to see the signal through the noise.”*

— **The Mountain Path Philosophy**

1 Multiple Linear Regression: A Quick Recap

Before we can understand why PCA is needed, we must first understand the regression framework it supports. Think of this section as laying the foundation of a building — every floor above depends on this being solid.

1.1 What Is Multiple Linear Regression (MLR)?

Definition: Multiple Linear Regression

MLR is a statistical method that models the relationship between **one** dependent variable (Y) and **two or more** independent variables ($X_1, X_2, \dots, X_p$). It extends Simple Linear Regression (which has only one X) to multiple predictors.

In plain English: “I want to predict Y (e.g., house price) using several inputs (X_1 = size, X_2 = bedrooms, X_3 = age, etc.) all at once.”

Why Do We Need Multiple Predictors?

In real life, outcomes are rarely driven by a single factor. A house’s price depends on size, location, age, number of rooms, garage, and more. Simple regression with just one X would miss most of the story. MLR lets us account for *all* relevant factors simultaneously.

1.2 The MLR Equation — Explained Piece by Piece

The standard MLR model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon$$

Let us break down every single term:

Symbol	Name	What It Means
Y	Response variable	The thing we are trying to predict (e.g., house price)
β_0	Intercept	The predicted value of Y when <i>all</i> X ’s are zero — the “baseline”
β_1	Coefficient of X_1	How much Y changes when X_1 increases by 1 unit, <i>holding all other X’s constant</i>
β_p	Coefficient of X_p	Same idea for the p -th predictor
ε	Error term	The part of Y we cannot explain — randomness, missing variables, measurement noise

Illustration

Concrete Example: Suppose we model house price as:

$$\text{Price} = 50,000 + 120 \times \text{Size(sqft)} + 15,000 \times \text{Bedrooms} - 2,000 \times \text{Age(yrs)} + \epsilon$$

Reading the coefficients:

- $\beta_0 = 50,000$: A hypothetical house with 0 sqft, 0 bedrooms, 0 age would cost \$50,000 (the baseline).
- $\beta_1 = 120$: Each additional square foot adds \$120 to the price, holding bedrooms and age fixed.
- $\beta_2 = 15,000$: Each additional bedroom adds \$15,000, holding size and age fixed.
- $\beta_3 = -2,000$: Each additional year of age *reduces* the price by \$2,000.

The phrase “holding others constant” is crucial — it means we are isolating the effect of each variable.

1.3 How Are the Coefficients (β) Estimated?

Ordinary Least Squares (OLS)

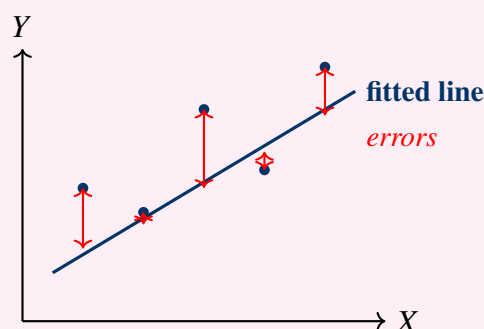
OLS is the most common method to estimate β values. The idea is simple: **choose the β 's that make the predicted values $\hat{Y}$ as close as possible to the actual values Y .**

Mathematically, we **minimise** the Sum of Squared Errors:

$$\text{SSE} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_{i1} - \dots - \beta_p X_{ip})^2$$

Why “Squared” Errors?

Why not just minimise the sum of errors? Because positive and negative errors would cancel out! Squaring ensures all errors are positive and penalises large errors more heavily than small ones.



The red vertical lines are the errors ($Y_i - \hat{Y}_i$). OLS makes the sum of their *squares* as small as possible.

The closed-form solution (using matrix algebra):

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

This formula requires us to **invert** the matrix $(\mathbf{X}^\top \mathbf{X})$. If this matrix is near-singular (almost

non-invertible), the estimates blow up. This is exactly what happens under multicollinearity — and exactly what PCA will fix.

1.4 The Five Key Assumptions of MLR

Assumptions of MLR — All Five Explained

- 1. Linearity** — The relationship between each X and Y is a straight line (not curved). If you plotted Y against any single X (holding others constant), you would see a linear trend.
- 2. Independence** — Each observation is collected independently. Knowing the value of one observation tells you nothing about another. (Violated in time-series data where today's value depends on yesterday's.)
- 3. Homoscedasticity** — The “spread” of errors is the same across all levels of X . Whether X is small or large, the errors have the same variance. The opposite (heteroscedasticity) means errors fan out like a trumpet.
- 4. Normality of errors** — The residuals ($Y - \hat{Y}$) follow a bell-shaped (normal) distribution. This is needed for valid p -values and confidence intervals.
- 5. No multicollinearity** — The predictor variables are **not** highly correlated with each other. Each X carries unique, independent information about Y .

Assumption 5 is the one that PCA is specifically designed to address. When predictors are correlated, the matrix $(\mathbf{X}^\top \mathbf{X})$ becomes nearly singular, and OLS breaks down. The rest of this guide is about understanding and solving this problem.

1.5 Model Evaluation Metrics

Metric	What It Tells You
R^2	Proportion of variance in Y explained by the model (ranges 0 to 1; higher = better fit). An R^2 of 0.85 means the model explains 85% of the variability in Y .
Adjusted R^2	Like R^2 but penalises adding useless variables. Prevents you from inflating R^2 by throwing in irrelevant predictors.
RMSE	Root Mean Squared Error — the typical size of your prediction mistake, in the same units as Y .
p -values	Tests whether each β is statistically different from zero. Small p -value (< 0.05) = the predictor is significant.

MLR works beautifully **when its assumptions hold**. The trouble starts when predictors are correlated with each other (multicollinearity). That is exactly where PCA enters the picture. The next section explains what goes wrong.

2 The Problem with MLR: Multicollinearity

2.1 What Is Multicollinearity?

Definition: Multicollinearity

Multicollinearity occurs when two or more predictor variables (X 's) in your MLR are **highly correlated** with each other — meaning they carry overlapping, redundant information.

Simple analogy: Imagine two witnesses in court telling the exact same story word-for-word. The judge cannot tell who actually saw the event and who is just repeating the other person. Similarly, when X_1 and X_2 move together, MLR cannot tell which one is truly driving Y .

Illustration

Real-world example: Predicting house price using:

- X_1 = area in square feet
- X_2 = number of rooms

These two variables move together — bigger houses have more rooms! The correlation between them might be $r = 0.92$. MLR struggles to separate their individual effects on price.

2.2 Why Is Multicollinearity a Problem? — Every Consequence Explained

The Five Consequences of Multicollinearity

1. Unstable coefficient estimates (β values swing wildly):

When predictors are correlated, adding or removing a single data point can cause β to change dramatically. For example, β_{size} might be +120 in one sample and −30 in another, even though size clearly increases price.

2. Inflated standard errors:

The standard error of $\hat{\beta}_j$ is proportional to $\sqrt{\text{VIF}_j}$. When VIF is large (due to multicollinearity), standard errors become huge, which makes t -statistics tiny, which makes p -values large. Result: statistically significant predictors *appear* insignificant.

3. Signs of coefficients may flip:

You might find $\beta_{\text{bedrooms}} < 0$, implying more bedrooms *decrease* price — nonsense! This happens because OLS is “confused” by the overlapping information and assigns credit arbitrarily.

4. The matrix $(\mathbf{X}^\top \mathbf{X})$ becomes near-singular:

Recall the OLS formula: $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$. If columns of $\mathbf{X}$ are nearly identical, the determinant of $(\mathbf{X}^\top \mathbf{X})$ approaches zero, making the inverse blow up. This is the mathematical root cause.

5. Overall R^2 stays high, but individual inference fails:

The model as a whole still predicts well (high R^2), but you cannot trust any *individual* coefficient. It is like having a team that wins games but you cannot tell which player contributed.

2.3 How Do We Detect Multicollinearity?

2.3.1 Method 1: Correlation Matrix

Compute the pairwise correlation coefficient between every pair of predictors. If $|r| > 0.8$, suspect multicollinearity.

What Is the Correlation Coefficient r ?

The correlation between two variables X and Z measures how strongly they move together:

$$r_{XZ} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Z_i - \bar{Z})}{\sqrt{\sum (X_i - \bar{X})^2} \cdot \sqrt{\sum (Z_i - \bar{Z})^2}}$$

Value of r	Interpretation
$r = +1$	Perfect positive linear relationship
$r = -1$	Perfect negative linear relationship
$r = 0$	No linear relationship
$ r > 0.8$	Strong — suspect multicollinearity

2.3.2 Method 2: Variance Inflation Factor (VIF)

VIF — The Gold Standard Detector

VIF measures how much the variance of $\hat{\beta}_j$ is **inflated** due to correlation with other predictors.

How to compute it:

1. Regress X_j on all the *other* predictors ($X_1, \dots, X_{j-1}, X_{j+1}, \dots, X_p$).
2. Get the R^2 from that regression (call it R_j^2).
3. Compute:
$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

Intuition: If X_j can be perfectly predicted by the other X 's ($R_j^2 = 1$), then $\text{VIF} = \infty$. If X_j is completely independent of the others ($R_j^2 = 0$), then $\text{VIF} = 1$.

VIF Value	Interpretation
VIF = 1	No multicollinearity at all
VIF 1–5	Moderate (usually acceptable)
VIF 5–10	Concerning (investigate further)
VIF > 10	Serious problem — must fix!

Illustration

Numerical example: Suppose we regress “Size” on Bedrooms, Age, Lot, and Garage, and get $R^2 = 0.994$. Then:

$$\text{VIF}_{\text{Size}} = \frac{1}{1 - 0.994} = \frac{1}{0.006} = 166.7$$

This is astronomically high! It means the standard error of $\hat{\beta}_{\text{Size}}$ is $\sqrt{166.7} \approx 12.9$ times larger than it would be if Size were uncorrelated with the other predictors.

2.3.3 Method 3: Eigenvalues Near Zero

If the eigenvalues of the correlation matrix include values very close to zero, it means the data lies in a lower-dimensional space — a hallmark of multicollinearity. (We will explain eigenvalues in full detail in Section 4.)

2.3.4 Method 4: Condition Number

The condition number is the ratio of the largest to smallest eigenvalue: $\kappa = \lambda_{\max} / \lambda_{\min}$. A value > 30 signals trouble.

2.4 The Usual Fixes (and Why PCA Is Special)

Method	Approach	Drawback
Drop variables	Remove one of the correlated pair	Loses information
Combine manually	Create ratios or composites	Arbitrary choices
Ridge Regression	Penalised regression; shrinks β	Keeps all variables (no reduction)
PCA	Mathematically build new uncorrelated predictors	Our focus!

Multicollinearity = correlated predictors = unstable β 's = unreliable inference. We can detect it with correlation matrices, VIF, or eigenvalues. PCA provides an elegant mathematical fix by *creating* new uncorrelated variables from the old correlated ones.

3 PCA — The Intuition (Before Any Math)

PCA in One Sentence

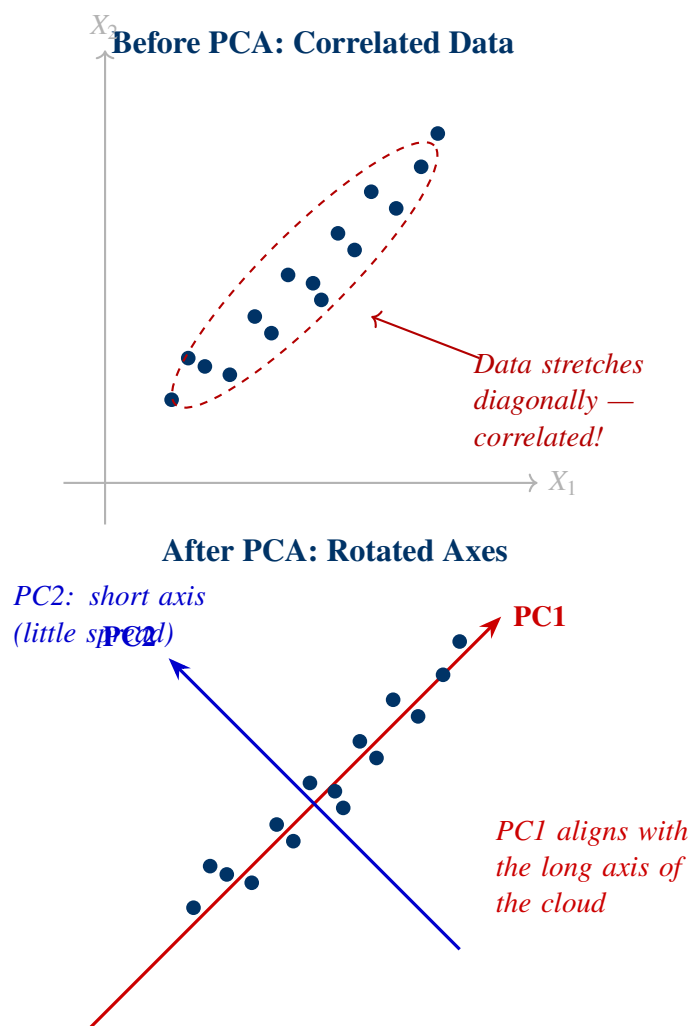
PCA is a technique that **transforms** your original correlated variables into a smaller set of **new, uncorrelated** variables (called **principal components**) that retain most of the information (variance) from the original data.

3.1 The Photography Analogy

Imagine you have a 3D sculpture and need to capture its essence in a single 2D photograph. From most angles, the photo looks flat and uninformative. But from one special angle, you see the maximum detail — the most “spread” or “variance” of the shape.

PCA does exactly this: it finds the “best viewing angles” for your data, mathematically.

3.2 The Diagonal Cloud Analogy



What Just Happened?

The data points did **not** move. We simply **rotated the coordinate axes** so that:

- **PC1** points along the direction of maximum spread (variance).
- **PC2** is perpendicular to PC1, capturing whatever spread remains.

In the original axes (X_1, X_2), the variables are correlated. In the new axes (PC1, PC2), they are **uncorrelated by construction**. This is the magic of PCA.

3.3 What Is a Principal Component?

Principal Component

A principal component is a **weighted sum** (linear combination) of the original variables:

$$PC_1 = w_{11} \cdot X_1 + w_{12} \cdot X_2 + \cdots + w_{1p} \cdot X_p$$

The weights (w 's) are called **loadings**. They tell you how much each original variable “contributes” to that PC. The loadings are chosen mathematically so that:

- PC1 captures the maximum possible variance.
- PC2 captures the maximum remaining variance, subject to being uncorrelated with PC1.
- And so on for PC3, PC4, etc.

3.4 The Three Magic Properties of Principal Components

Property 1: Orthogonal (Zero Correlation)

All PCs are perpendicular (orthogonal) to each other. This means:

$$\text{Corr}(PC_i, PC_j) = 0 \quad \text{for all } i \neq j$$

Why this matters: This is exactly the property that cures multicollinearity. When we use PCs as predictors in MLR, the predictors are guaranteed to be uncorrelated!

Property 2: Ordered by Variance

PCs are numbered in decreasing order of variance:

$$\text{Var}(PC_1) \geq \text{Var}(PC_2) \geq \cdots \geq \text{Var}(PC_p)$$

PC1 is always the “most informative” direction. PC_p (the last one) captures the least information and is often just noise.

Property 3: Total Variance Is Preserved

PCA does not destroy information. The total variance across all PCs equals the total variance across all original variables:

$$\text{Var}(\text{PC}_1) + \text{Var}(\text{PC}_2) + \cdots + \text{Var}(\text{PC}_p) = \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_p)$$

PCA merely **redistributes** variance: it packs as much as possible into the first few PCs, allowing us to drop the rest with minimal loss.

PCA = rotation of axes to align with directions of greatest spread. It does not delete data — it reorganises it. The new variables (PCs) are uncorrelated and ordered by importance. This is exactly what we need to fix multicollinearity before running MLR.

4 The Math Behind PCA — A Deep Dive

This is the heart of the guide. We will explain every mathematical concept from scratch, with examples.

4.1 Step 1: The Mean — Foundation of Everything

What Is the Mean?

The mean (average) of a variable X with n observations is:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

The mean is the “centre of gravity” of the data. Centring means subtracting the mean from each value: $X_i - \bar{X}$. After centring, the data has mean zero.

4.2 Step 2: Variance — Measuring Spread

What Is Variance?

Variance measures how far data points are spread out from their mean:

$$\text{Var}(X) = s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

We divide by $(n-1)$ instead of n for the “sample variance” (Bessel’s correction).

Intuition: If all data points are identical, variance = 0. The more spread out they are, the larger the variance.

Worked Calculation: Variance of X_1

Using our 10-observation dataset from Section 7 (raw values: 2.5, 0.5, 2.2, 1.9, 3.1, 2.3, 2.0, 1.0, 1.5, 1.1):

Step A: Mean = $\bar{X}_1 = (2.5 + 0.5 + 2.2 + \cdots + 1.1)/10 = 18.1/10 = 1.81$

Step B: Deviations ($X_i - \bar{X}$): +0.69, −1.31, +0.39, +0.09, +1.29, +0.49, +0.19, −0.81, −0.31, −0.71

Step C: Squared deviations: 0.4761, 1.7161, 0.1521, 0.0081, 1.6641, 0.2401, 0.0361, 0.6561, 0.0961, 0.5041

Step D: Sum = 5.5490

Step E: $\text{Var}(X_1) = 5.5490/(10-1) = 5.5490/9 = 0.6166$

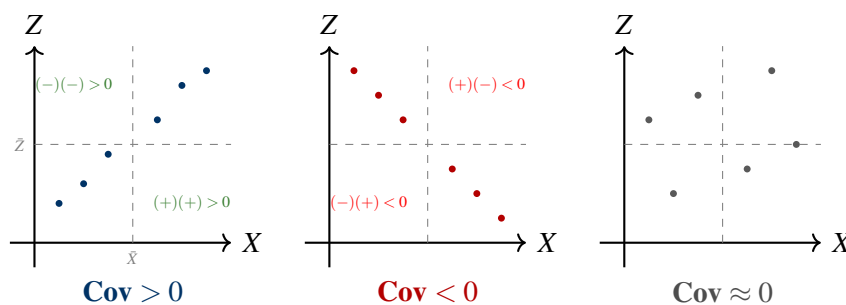
4.3 Step 3: Covariance — How Two Variables Move Together

What Is Covariance?

Covariance measures whether two variables tend to increase together (positive), move in opposite directions (negative), or have no linear relationship (near zero):

$$\text{Cov}(X, Z) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Z_i - \bar{Z})$$

Key insight: When both deviations have the *same* sign (both above or both below their means), the product is positive. When they have *opposite* signs, the product is negative. Summing these products tells us the overall tendency.



4.4 Step 4: The Covariance Matrix Σ

What Is the Covariance Matrix?

For p variables, the covariance matrix is a $p \times p$ symmetric matrix where:

- **Diagonal entries** = variance of each variable: $\Sigma_{jj} = \text{Var}(X_j)$
- **Off-diagonal entries** = covariance between pairs: $\Sigma_{jk} = \text{Cov}(X_j, X_k)$

For two variables ($p = 2$):

$$\Sigma = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) \end{pmatrix}$$

Note: $\text{Cov}(X_1, X_2) = \text{Cov}(X_2, X_1)$, so the matrix is symmetric.

Our 2-Variable Covariance Matrix (from the Worked Example)

Using the 10-observation dataset (X_1, X_2) :

$$\Sigma = \begin{pmatrix} 0.6166 & 0.6154 \\ 0.6154 & 0.7166 \end{pmatrix}$$

Reading it:

- $\Sigma_{11} = 0.6166 = \text{variance of } X_1$
- $\Sigma_{22} = 0.7166 = \text{variance of } X_2$
- $\Sigma_{12} = \Sigma_{21} = 0.6154 = \text{covariance of } X_1 \text{ and } X_2 \text{ (strongly positive } \Rightarrow \text{ they move together)}$

PCA's goal: find a rotation that **diagonalises** this matrix — meaning the off-diagonal entries become zero (no correlation).

4.5 Step 5: Eigenvalues and Eigenvectors — The Core of PCA

This is the most important mathematical concept in PCA. We will build it from scratch.

4.5.1 What Is an Eigenvector? What Is an Eigenvalue?

Definition: Eigenvalue and Eigenvector

For a square matrix $\mathbf{A}$, if there exists a non-zero vector $\mathbf{v}$ and a scalar λ such that:

$$\mathbf{A} \cdot \mathbf{v} = \lambda \cdot \mathbf{v}$$

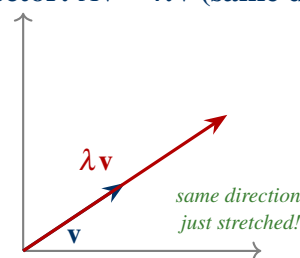
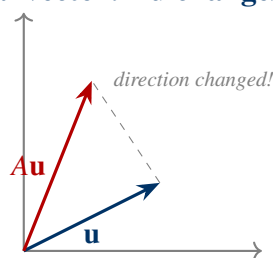
then $\mathbf{v}$ is called an **eigenvector** of $\mathbf{A}$, and λ is the corresponding **eigenvalue**.

What Does This Equation Mean

Normally, multiplying a matrix $\mathbf{A}$ by a vector $\mathbf{v}$ changes *both* the direction and length of $\mathbf{v}$. But for special vectors (eigenvectors), the matrix only **stretches or compresses** them — their direction stays the same!

- **Eigenvector** $\mathbf{v}$ = a special direction that the matrix does not rotate, only stretches.
- **Eigenvalue** λ = the stretching factor. $\lambda = 2$ means the vector gets doubled in length. $\lambda = 0.5$ means it gets halved.

General vector: $\mathbf{A}\mathbf{u}$ changes direction. **Eigenvector:** $\mathbf{A}\mathbf{v} = \lambda \mathbf{v}$ (same direction)



4.5.2 Why Do Eigenvalues Matter for PCA?

The Connection to PCA

When we compute eigenvalues and eigenvectors of the **covariance matrix** Σ :

- Each **eigenvector** tells us a direction of a principal component (a new axis).
- Each **eigenvalue** tells us how much variance (information) lies along that direction.
- The largest eigenvalue λ_1 corresponds to PC1 (the direction of maximum variance).
- The smallest eigenvalue λ_p corresponds to the direction of least variance (often just noise).

Summary: Eigenvectors = directions of the new axes. Eigenvalues = how much data spread (importance) is in each direction.

4.5.3 How to Calculate Eigenvalues — The General Method

Step-by-Step: Finding Eigenvalues

Starting from $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, rearrange:

$$\mathbf{A}\mathbf{v} - \lambda\mathbf{v} = \mathbf{0} \quad \Rightarrow \quad (\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$$

where $\mathbf{I}$ is the identity matrix.

For a non-zero solution $\mathbf{v}$ to exist, the matrix $(\mathbf{A} - \lambda\mathbf{I})$ must be **singular** (non-invertible).

This happens when:

$$\det(\mathbf{A} - \lambda\mathbf{I}) = 0$$

This equation is called the **characteristic equation**. Solving it gives us the eigenvalues λ .

4.5.4 The Closed-Form Solution for a 2×2 Matrix

For a 2×2 matrix, we can solve the characteristic equation using the quadratic formula. Let us work through this step-by-step.

Eigenvalues of a 2×2 Matrix — Complete Derivation

Consider a general 2×2 matrix:

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

Step 1: Set up the characteristic equation.

$$\mathbf{A} - \lambda \mathbf{I} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix}$$

Step 2: Compute the determinant and set it to zero.

$$\det(\mathbf{A} - \lambda \mathbf{I}) = (a - \lambda)(d - \lambda) - bc = 0$$

Step 3: Expand.

$$ad - a\lambda - d\lambda + \lambda^2 - bc = 0$$

$$\lambda^2 - (a + d)\lambda + (ad - bc) = 0$$

Step 4: Recognise the terms.

- $a + d$ = the **Trace** of the matrix (sum of diagonal elements).
- $ad - bc$ = the **Determinant** of the matrix.

So the characteristic equation is:

$$\lambda^2 - \text{Trace} \cdot \lambda + \text{Det} = 0$$

Step 5: Apply the quadratic formula.

For $ax^2 + bx + c = 0$, the solution is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Here: $a = 1$, $b = -\text{Trace}$, $c = \text{Det}$:

$$\lambda = \frac{\text{Trace} \pm \sqrt{\text{Trace}^2 - 4 \cdot \text{Det}}}{2}$$

The term under the square root, $\text{Trace}^2 - 4 \cdot \text{Det}$, is called the **Discriminant**.

4.5.5 Complete Numerical Calculation with Our Data

Eigenvalue Calculation — Our 2-Variable Example

Our covariance matrix is:

$$\Sigma = \begin{pmatrix} 0.6166 & 0.6154 \\ 0.6154 & 0.7166 \end{pmatrix}$$

where $a = 0.6166$, $b = 0.6154$, $c = 0.6154$, $d = 0.7166$.

Step 1: Compute the Trace.

$$\text{Trace} = a + d = 0.6166 + 0.7166 = 1.3332$$

(Sum of the diagonal — total variance across both variables.)

Step 2: Compute the Determinant.

$$\begin{aligned} \text{Det} &= ad - bc = (0.6166)(0.7166) - (0.6154)(0.6154) \\ &= 0.4417 - 0.3787 = 0.0630 \end{aligned}$$

(How “non-singular” the matrix is. Small Det means high correlation.)

Step 3: Compute the Discriminant.

$$\begin{aligned} \text{Disc} &= \text{Trace}^2 - 4 \times \text{Det} = (1.3332)^2 - 4(0.0630) \\ &= 1.7774 - 0.2520 = 1.5254 \\ \sqrt{\text{Disc}} &= \sqrt{1.5254} = 1.2351 \end{aligned}$$

Step 4: Apply the quadratic formula.

$$\begin{aligned} \lambda_1 &= \frac{\text{Trace} + \sqrt{\text{Disc}}}{2} = \frac{1.3332 + 1.2351}{2} = \frac{2.5683}{2} = 1.2841 \\ \lambda_2 &= \frac{\text{Trace} - \sqrt{\text{Disc}}}{2} = \frac{1.3332 - 1.2351}{2} = \frac{0.0981}{2} = 0.0491 \end{aligned}$$

Quantity	Formula	Our Value
Trace ($a + d$)	Sum of diagonal	1.3332
Determinant ($ad - bc$)	Product of diag – product of off-diag	0.0630
Discriminant	$\text{Trace}^2 - 4 \times \text{Det}$	1.5254
$\sqrt{\text{Disc}}$		1.2351
λ_1 (largest)	$(\text{Trace} + \sqrt{\text{Disc}})/2$	1.2841
λ_2 (smallest)	$(\text{Trace} - \sqrt{\text{Disc}})/2$	0.0491

Check: $\lambda_1 + \lambda_2 = 1.2841 + 0.0491 = 1.3332 = \text{Trace} \checkmark$

(The sum of eigenvalues always equals the trace — the total variance is preserved!)

4.5.6 How to Calculate Eigenvectors

Finding the Eigenvector for λ_1

We solve $(\Sigma - \lambda_1 \mathbf{I})\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} 0.6166 - 1.2841 & 0.6154 \\ 0.6154 & 0.7166 - 1.2841 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -0.6675 & 0.6154 \\ 0.6154 & -0.5675 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

From the first row: $-0.6675 v_1 + 0.6154 v_2 = 0$

$$v_2 = \frac{0.6675}{0.6154} v_1 = 1.0846 v_1$$

Choose $v_1 = 1$, so $v_2 = 1.0846$. Now **normalise** (make length = 1):

$$\|\mathbf{v}\| = \sqrt{1^2 + 1.0846^2} = \sqrt{1 + 1.1764} = \sqrt{2.1764} = 1.4753$$

$$\mathbf{v}_1 = \left(\frac{1}{1.4753}, \frac{1.0846}{1.4753} \right) = (0.6779, 0.7352)$$

This is the PC1 direction! Both entries are positive and similar in size, meaning PC1 represents “overall size” — when both X_1 and X_2 are large, PC1 is large.

Finding the Eigenvector for λ_2

Similarly, solve $(\Sigma - \lambda_2 \mathbf{I})\mathbf{v} = \mathbf{0}$:

From the first row: $(0.6166 - 0.0491)v_1 + 0.6154 v_2 = 0$, so $0.5675 v_1 + 0.6154 v_2 = 0$

$$v_2 = -\frac{0.5675}{0.6154} v_1 = -0.9222 v_1$$

Choose $v_1 = 1$, $v_2 = -0.9222$. Normalise:

$$\|\mathbf{v}\| = \sqrt{1 + 0.8505} = \sqrt{1.8505} = 1.3603$$

$$\mathbf{v}_2 = \left(\frac{1}{1.3603}, \frac{-0.9222}{1.3603} \right) = (0.7352, -0.6779)$$

This is the PC2 direction! The signs are *opposite*, meaning PC2 captures the *contrast* between X_1 and X_2 (when X_1 is high but X_2 is low, or vice versa).

Verification: Are the Eigenvectors Perpendicular?

Two vectors are perpendicular (orthogonal) if their dot product = 0:

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = (0.6779)(0.7352) + (0.7352)(-0.6779) = 0.4984 - 0.4984 = 0 \quad \checkmark$$

They are indeed perpendicular. This confirms our PCs are uncorrelated.

4.6 Step 6: Explained Variance — How Important Is Each PC?

Explained Variance Ratio

The proportion of total variance captured by PC_i is:

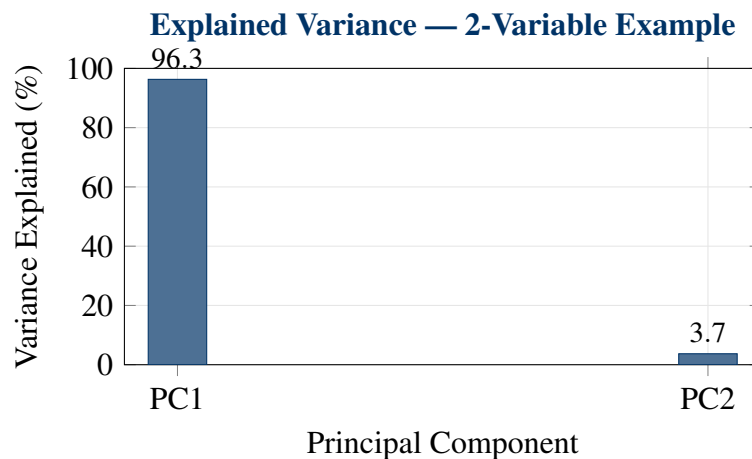
$$\text{Explained Variance Ratio}_i = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \dots + \lambda_p}$$

Calculation for Our Example

Total variance = $\lambda_1 + \lambda_2 = 1.2841 + 0.0491 = 1.3332$

$$PC1 \% = \frac{1.2841}{1.3332} = 96.32\% \quad PC2 \% = \frac{0.0491}{1.3332} = 3.68\%$$

PC1 alone captures 96.3% of all the information! We could drop PC2 and lose only 3.7%.



4.7 Step 7: Computing PC Scores

What Are PC Scores?

PC scores are the **coordinates** of each data point in the new PC coordinate system. They are computed by projecting the centred data onto the eigenvectors:

$$PC1 \text{ score}_i = (X_{i1} - \bar{X}_1) \times v_{11} + (X_{i2} - \bar{X}_2) \times v_{12}$$

$$PC2 \text{ score}_i = (X_{i1} - \bar{X}_1) \times v_{21} + (X_{i2} - \bar{X}_2) \times v_{22}$$

In matrix form: $\mathbf{Z} = \mathbf{X}_{\text{centred}} \cdot \mathbf{V}$

Worked Calculation: PC Scores for Observation 1

Observation 1: $X_1 = 2.5, X_2 = 2.4$

Centred values: $X_1 - \bar{X}_1 = 2.5 - 1.81 = 0.69, \quad X_2 - \bar{X}_2 = 2.4 - 1.91 = 0.49$

$$PC1 = (0.69)(0.6779) + (0.49)(0.7352) = 0.4678 + 0.3602 = 0.828$$

$$PC2 = (0.69)(0.7352) + (0.49)(-0.6779) = 0.5073 - 0.3322 = 0.175$$

Interpretation: This house is above average on PC1 (“overall size”) and slightly above average on PC2 (“ X_1 relatively higher than X_2 ”).

Sanity Check: Are PC Scores Uncorrelated?

Computing the correlation between all PC1 scores and all PC2 scores gives:

$$\text{Corr}(PC1, PC2) \approx 0.0000$$

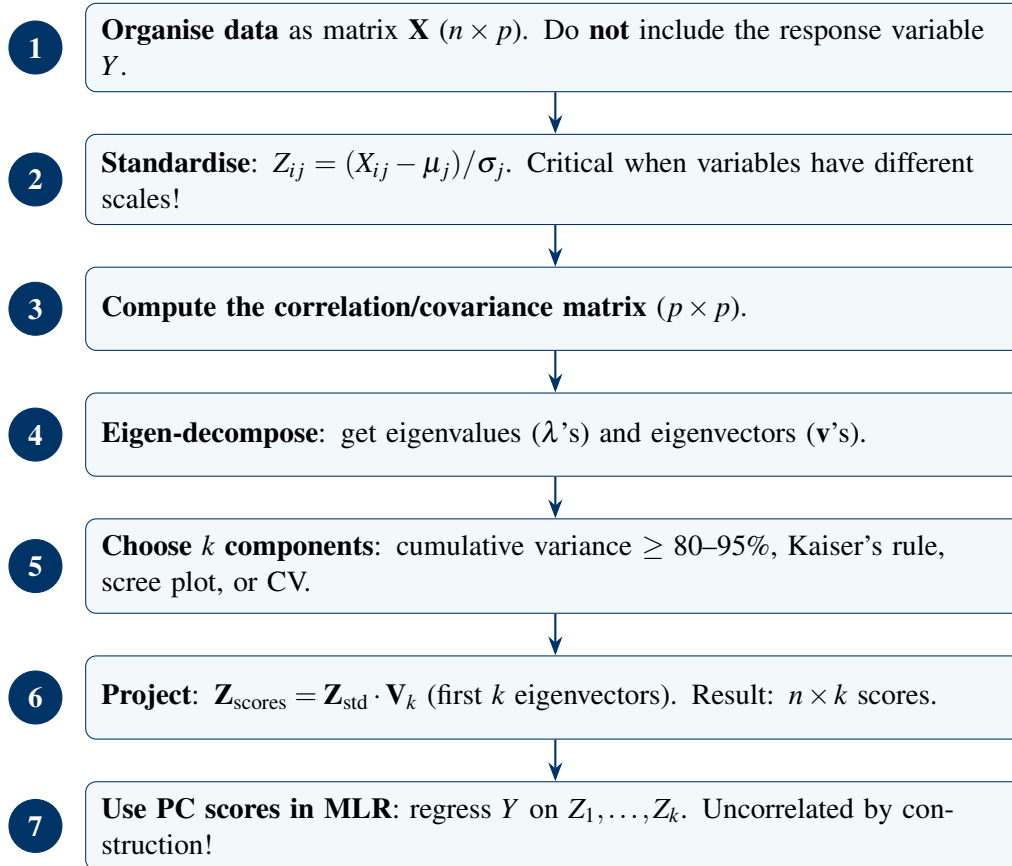
This confirms: the new variables (PC scores) are **uncorrelated**, just as the theory promises. This is what makes them safe to use in MLR.

Input: A covariance matrix ($p \times p$) summarising relationships among variables.

Process: Eigen-decomposition finds special directions (eigenvectors) and their importance (eigenvalues).

Output: New uncorrelated variables (PC scores) ordered from most to least informative. The first few PCs usually capture $>90\%$ of the variance, letting us discard the rest with minimal loss.

5 PCA Step-by-Step Recipe



5.1 Choosing the Number of Components (k) — Four Methods Explained

Method 1: Cumulative Variance Rule

Keep adding PCs until the cumulative explained variance reaches a threshold (typically 80–95%):

$$\frac{\lambda_1 + \lambda_2 + \dots + \lambda_k}{\lambda_1 + \lambda_2 + \dots + \lambda_p} \geq \text{threshold}$$

Example: If 3 PCs explain 99.2% of variance (as in our case study), keeping $k = 3$ out of 5 is well justified.

Method 2: Kaiser's Rule

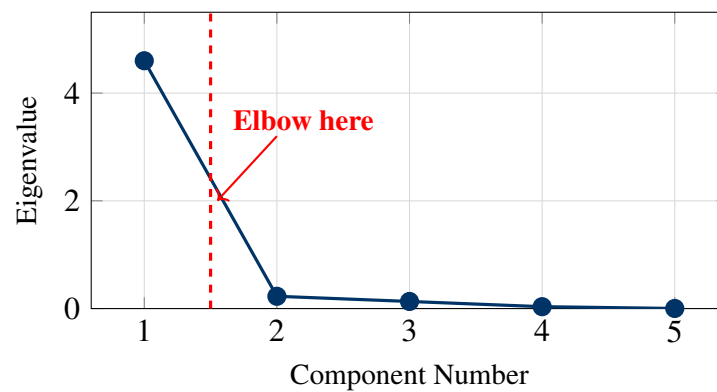
Keep PCs with eigenvalue > 1 (when using the *correlation* matrix).

Logic: If the correlation matrix is used, each standardised variable has variance = 1. A PC with $\lambda > 1$ captures more information than any single original variable — worth keeping!

Method 3: Scree Plot

Plot eigenvalues on a bar chart and look for the “elbow” — the point where eigenvalues drop sharply and then flatten out. Keep the PCs *before* the elbow.

Scree Plot — Case Study (5 Variables)

**Method 4: Cross-Validation (Best for Prediction)**

For PCR, try different values of k (1, 2, 3, ...) and evaluate prediction error on a held-out validation set. Choose the k that minimises prediction error. This is the most rigorous approach but requires more data.

Critical Pitfall: Forgetting to Standardise!

If you run PCA on unstandardised data, variables with the largest numerical scale will dominate PC1 regardless of their true importance. A “Lot size (sqft)” variable in the thousands will swamp a “Garage” variable in single digits purely because of units, not because lot size carries more information. **Always standardise when variables are on different scales!**

6 Principal Component Regression (PCR)

What Is PCR?

PCR = PCA followed by MLR. Instead of regressing Y on the original (correlated) predictors $X_1, \dots, X_p$, we regress Y on the (uncorrelated) principal component scores $Z_1, \dots, Z_k$.

6.1 The PCR Equation — Explained

$$Y = \alpha_0 + \alpha_1 Z_1 + \alpha_2 Z_2 + \dots + \alpha_k Z_k + \varepsilon$$

Symbol	Name	Meaning
$Z_1, \dots, Z_k$	PC scores	New uncorrelated predictors (from PCA)
α_j	PC coefficient	How much Y changes per unit increase in PC_j
k	Number of PCs kept	Chosen by variance threshold, scree plot, or CV
ε	Error term	Unexplained variation

6.2 Why Does PCR Cure Multicollinearity?

The Mathematical Reason

Because PCs are orthogonal (uncorrelated), the matrix $(\mathbf{Z}^\top \mathbf{Z})$ is **diagonal**:

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & & \ddots & \\ 0 & 0 & \dots & \lambda_k \end{pmatrix}$$

A diagonal matrix is trivially invertible (just invert each diagonal entry):

$$(\mathbf{Z}^\top \mathbf{Z})^{-1} = \begin{pmatrix} 1/\lambda_1 & 0 & \dots \\ 0 & 1/\lambda_2 & \dots \\ \vdots & & \ddots \end{pmatrix}$$

No near-zero determinant, no explosion, no inflated standard errors. Each α_j is estimated independently and stably.

6.3 Translating Back to Original Variables

After fitting PCR, you can express the results in terms of the original X 's:

$$\hat{\beta} = \mathbf{V}_k \cdot \hat{\alpha}$$

where $\mathbf{V}_k$ contains the loadings (eigenvectors). If you standardised X , also rescale by the original standard deviations.

6.4 PCR Workflow Cheat Sheet

PCR in 6 Steps

1. **Standardise X:** $Z = (X - \mu)/\sigma$
2. **PCA:** eigen-decompose the correlation matrix; get eigenvectors $\mathbf{V}$ and eigenvalues λ
3. **Choose k :** pick the number of PCs to keep
4. **Compute scores:** $\mathbf{Z}_{\text{scores}} = \mathbf{Z}_{\text{std}} \times \mathbf{V}_k$
5. **Fit OLS:** $Y = \alpha_0 + \alpha_1 Z_1 + \dots + \alpha_k Z_k$
6. **Predict:** for new $\mathbf{x}$, standardise with *training* μ, σ , project onto *training* $\mathbf{V}_k$, plug into fitted equation

6.5 PCR vs. Ridge vs. PLS — When to Use What

Method	How It Works	Key Characteristic	Best When
PCR	Transforms X using PCA (ignores Y). Drops low-variance PCs.	Unsupervised dimensionality reduction	Prediction > interpretation
Ridge	Adds penalty $\lambda \ \beta\ ^2$ to OLS. Keeps all variables but shrinks coefficients.	Supervised regularisation	Want original-variable coefficients
PLS	Builds components using <i>both</i> X -variance and Y -correlation.	Supervised reduction	Want components that predict Y well

Important Caveat

PC1 captures the most variance in X , but that does **not** guarantee it best predicts Y . Sometimes a low-variance PC carries the signal relevant to Y . Always validate with cross-validation!

7 Worked Illustration: Complete 2-Variable PCA

This section consolidates everything from Section 4 into a single end-to-end example.

Scenario

Two predictors X_1 and X_2 are highly correlated (correlation ≈ 0.93). We perform PCA step-by-step on 10 observations.

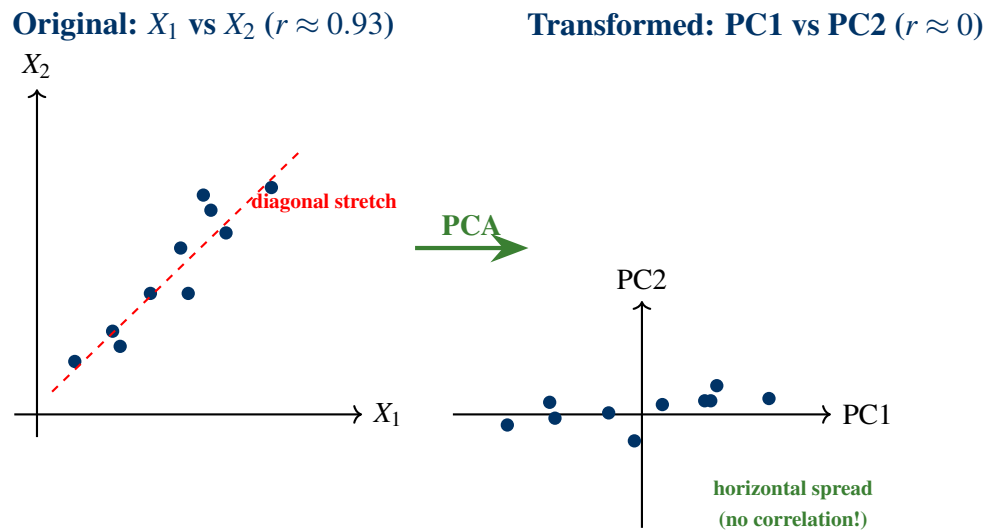
7.1 The Complete Data Table

Obs	X_1	X_2	$X_1 - \bar{X}_1$	$X_2 - \bar{X}_2$	PC1 Score	PC2 Score
1	2.5	2.4	+0.69	+0.49	+0.828	+0.175
2	0.5	0.7	-1.31	-1.21	-1.778	-0.143
3	2.2	2.9	+0.39	+0.99	+0.992	+0.384
4	1.9	2.2	+0.09	+0.29	+0.274	+0.130
5	3.1	3.0	+1.29	+1.09	+1.676	+0.210
6	2.3	2.7	+0.49	+0.79	+0.913	+0.175
7	2.0	1.6	+0.19	-0.31	-0.099	-0.350
8	1.0	1.1	-0.81	-0.81	-1.145	-0.046
9	1.5	1.6	-0.31	-0.31	-0.438	+0.018
10	1.1	0.9	-0.71	-1.01	-1.224	+0.163
Mean	1.81	1.91	0.00	0.00	0.000	0.000

7.2 Summary of All Calculations

	PC1	PC2
Eigenvalue λ	1.2841	0.0491
Variance %	96.32%	3.68%
Cumulative %	96.32%	100.00%
Loading on X_1	+0.678	+0.735
Loading on X_2	+0.735	-0.678
Interpretation	“Overall size”	“ X_1 vs X_2 contrast”
Corr(PC1, PC2)	≈ 0.000 (uncorrelated!)	

7.3 Visual: Before vs. After PCA



The diagonal stretch (correlation) in the original data has been “unrotated” into a pure horizontal spread along PC1. The cloud now has almost no vertical spread ($PC2 \approx \text{noise}$). With one PC instead of two correlated X ’s, MLR becomes simpler, stabler, and just as informative.

8 Pros, Cons, and Pitfalls of PCR

Advantages	Disadvantages
Removes multicollinearity completely — PCs are orthogonal	PCs are abstract mathematical combinations; harder to interpret than original variables
Reduces dimensionality — fewer predictors, less overfitting	Unsupervised — PCA ignores Y ; high-variance PCs may not predict Y well
Numerical stability — $(\mathbf{Z}^\top \mathbf{Z})$ is diagonal, easy to invert	Sensitive to scaling — forgetting to standardise gives misleading components
Noise filtering — dropping low-variance PCs often removes noise	Assumes linearity — PCA only captures linear structure
Visualisation — first 2–3 PCs let you plot high-dimensional data	Sensitive to outliers — a single extreme point can dominate PC directions
Works on any number of predictors, including $p > n$ cases	Choosing k involves judgement — different rules give different answers

8.1 Five Common Pitfalls — Explained in Detail

Pitfall 1: Forgetting to Standardise

What happens: The variable with the largest numerical scale dominates PC1. A “Lot size” in thousands of sqft will swamp “Garage spaces” in single digits.

Fix: Always compute $Z = (X - \mu)/\sigma$ before PCA when variables have different units or scales.

Pitfall 2: Including Y in the PCA

What happens: PCA is an *unsupervised* method — it should only see the predictor variables (X ’s). If you include Y , you contaminate the components with the response, leading to overfitting and data leakage.

Fix: Run PCA on X only. Regress Y on the resulting PC scores *afterwards*.

Pitfall 3: Always Keeping Just PC1

What happens: Sometimes a “small” PC (say PC3 with only 2% variance) carries critical predictive power for Y . If you blindly keep only PC1, you lose that signal.

Fix: Use cross-validation to choose k , not just a variance threshold.

Pitfall 4: Interpreting PC Coefficients as Causal Effects

What happens: PC1 is a *direction* in variable space, not a single feature. Saying “PC1 causes higher prices” is meaningless. PCs are mathematical constructs, not causal drivers.

Fix: Use loadings to translate back to original variables for business interpretation.

Pitfall 5: Recomputing PCA on New Data

What happens: When predicting for a new observation, you must reuse the *training* set's μ , σ , and eigenvectors $\mathbf{V}$. If you recompute PCA on the new data, the coordinate system changes and predictions become meaningless.

Fix: Store μ , σ , and $\mathbf{V}$ from training. For new $\mathbf{x}$: standardise with training μ, σ ; project with training $\mathbf{V}_k$.

9 Glossary and FAQ

9.1 Glossary of Key Terms

Term	Definition
Multicollinearity	High correlation among predictor variables, causing unstable regression coefficients
Principal Component	Linear combination of original variables; a new uncorrelated axis
Loading	Weight of an original variable in a PC (entry of the eigenvector)
Score	Value of a PC for a specific observation (projection onto eigenvector)
Eigenvalue (λ)	Variance captured by its corresponding PC direction
Eigenvector ($\mathbf{v}$)	Direction in feature space that defines a PC; the covariance matrix only stretches (does not rotate) this vector
Trace	Sum of diagonal elements of a matrix; equals the sum of all eigenvalues
Determinant	Product of eigenvalues; measures how “invertible” a matrix is
Characteristic Equation	$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$; solving this gives the eigenvalues
Explained Variance	$\lambda_i / \sum \lambda$; fraction of total variance captured by PC_i
Scree Plot	Plot of eigenvalues vs. component number; used to choose k (look for the elbow)
Standardisation	Transform to mean 0, std 1: $Z = (X - \mu) / \sigma$
VIF	Variance Inflation Factor; $= 1 / (1 - R_j^2)$; measures multicollinearity impact
PCR	Principal Component Regression; MLR using PC scores as predictors
PLS	Partial Least Squares; like PCR but components also account for Y correlation
SVD	Singular Value Decomposition; numerically stable method to compute PCA

9.2 Frequently Asked Questions

Q1: Should I always standardise before PCA?

Almost always **YES**, especially when variables are on different scales or units. If all variables are already on a comparable scale (e.g., all percentages), centring alone may suffice. When in doubt, standardise.

Q2: How many PCs should I keep?

Use one or more of: cumulative explained variance ($\sim 80\text{--}95\%$), scree-plot elbow, Kaiser's rule ($\lambda > 1$), or cross-validation. For PCR, CV with respect to prediction error is the most defensible.

Q3: Can I interpret PCs in terms of original variables?

Yes, by examining the **loadings** (eigenvector entries). Large absolute loadings indicate strong contribution. Often a PC represents a meaningful concept (e.g., "overall size", "age vs. size contrast").

Q4: Is PCR always better than plain MLR?

No. If predictors are not multicollinear and are easily interpretable, plain MLR is preferable. PCR shines under multicollinearity, high p , or when prediction matters more than interpretation.

Q5: Does PCA work for categorical variables?

Standard PCA is for **continuous numeric data**. For categorical data, use Multiple Correspondence Analysis (MCA) or one-hot encode first (with caution regarding dimensionality).

Q6: How do I predict for a new observation in PCR?

Centre and scale $\mathbf{x}_{\text{new}}$ using the **training** μ and σ . Multiply by training $\mathbf{V}_k$. Plug the resulting PC scores into the fitted MLR equation. **Never** recompute PCA on the new point.

10 Case Study: Predicting House Prices with PCR

Business Problem

A real-estate analyst wants to predict house price (Y) using 5 features: Size (sqft), Bedrooms, Age (years), Lot size (sqft), and Garage spaces. Many features move together — bigger houses have more bedrooms, larger lots, and more garages. **Multicollinearity is highly likely.**

10.1 Step 1: The Raw Data ($n = 12$ houses, $p = 5$ features)

House	Size	Beds	Age	Lot	Garage	Price (\$k)
1	1500	3	20	5000	1	245
2	2200	4	5	7500	2	380
3	1800	3	15	6000	2	295
4	2500	4	8	8200	2	425
5	1200	2	35	4200	1	180
6	3000	5	3	10000	3	540
7	1700	3	25	5500	1	265
8	2800	4	10	9000	2	475
9	2000	3	18	6500	2	320
10	1400	2	30	4800	1	210
11	2600	4	7	8500	3	455
12	1900	3	22	6200	2	305

10.2 Step 2: Detect the Problem — The Correlation Matrix

	Size	Beds	Age	Lot	Garage
Size	1.000	0.942	−0.896	red!150.997	0.858
Beds	0.942	1.000	red!15−0.946	0.951	0.809
Age	−0.896	−0.946	1.000	−0.907	−0.825
Lot	red!150.997	0.951	−0.907	1.000	0.866
Garage	0.858	0.809	−0.825	0.866	1.000

Red Flags Everywhere!

Size ↔ Lot: $r = 0.997$ (almost perfectly correlated!)

Beds ↔ Age: $r = -0.946$

Size ↔ Beds: $r = 0.942$

Almost every pair has $|r| > 0.8$. This is classic, severe multicollinearity. Plain MLR coefficients will be wildly unstable.

10.3 Step 3: Standardise All Predictors

Why Standardise? A Concrete Reason

Look at the raw scales: Size ranges from 1200–3000, Lot from 4200–10000, but Garage is just 1–3. Without standardisation, PCA would be completely dominated by Lot (largest numbers), even though Garage might be equally informative. Standardising puts every variable on a level playing field: mean = 0, std = 1.

Statistic	Size	Beds	Age	Lot	Garage
Mean (μ)	2050	3.33	16.5	6783	1.83
Std Dev (σ)	571.5	0.888	10.01	1822	0.718

Each Z-score: $Z_{ij} = (X_{ij} - \mu_j) / \sigma_j$. After this, every column has mean = 0 and std = 1.

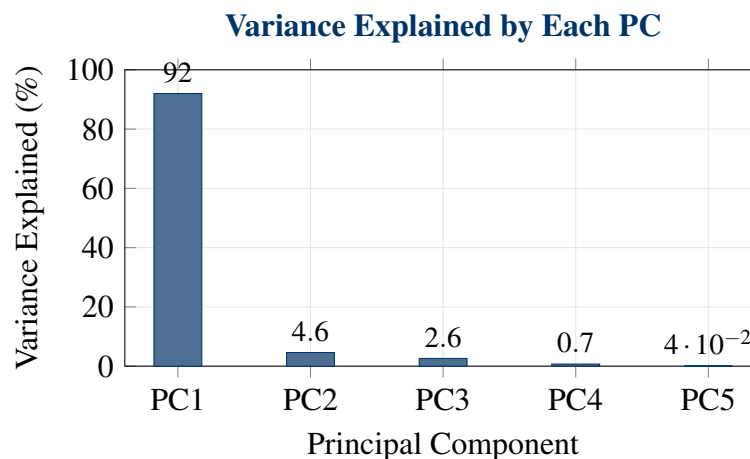
10.4 Step 4: Eigen-Decomposition Results

PC	Eigenvalue (λ)	Variance %	Cumulative %
PC1	4.602	92.0%	92.0%
PC2	0.228	4.6%	96.6%
PC3	0.132	2.6%	99.2%
PC4	0.035	0.7%	99.9%
PC5	0.002	0.04%	100.0%

Reading the Eigenvalue Table

Why is $\lambda_1 = 4.602$ so dominant? Because the 5 variables are so highly correlated, they are essentially measuring the same underlying concept (“how big and new is the house?”). One direction (PC1) captures almost all the variation. The remaining eigenvalues are tiny because there is very little independent information left after accounting for that one direction.

Decision: PC1 + PC2 + PC3 = 99.2%. We keep $k = 3$ components and discard PC4 and PC5 (which together account for only 0.8% of variance).



10.5 Step 5: Loadings — Interpreting What Each PC Means

Variable	PC1	PC2	PC3
Size	+0.457	−0.083	+0.517
Bedrooms	+0.452	−0.391	−0.115
Age	−0.445	+0.265	+0.702
Lot	+0.459	−0.079	+0.442
Garage	+0.422	+0.874	−0.176

Illustration

How to read loadings:

PC1 has loadings of similar magnitude (~ 0.45) on Size, Bedrooms, Lot, and Garage (all positive), and a negative loading on Age (-0.445). This means:

- When a house is big, has many bedrooms, a large lot, multiple garages, and is *young* — PC1 is high.
- **Interpretation: PC1 = “Overall house bigness and newness.”**

PC2 is dominated by Garage ($+0.874$). All other loadings are small. This means:

- PC2 mainly captures variation in garage count that is *not* explained by overall size.
- **Interpretation: PC2 = “Extra garage capacity beyond what size predicts.”**

PC3 has strong loadings on Age ($+0.702$) and Size ($+0.517$):

- PC3 captures houses that are *both old and large* (or both new and small).
- **Interpretation: PC3 = “Old but large houses” (a niche combination).**

10.6 Step 6: PCR Results — Regression on PC Scores

Regressing Price on PC1, PC2, and PC3:

	$\alpha(\text{PC1})$	$\alpha(\text{PC2})$	$\alpha(\text{PC3})$	Intercept
Estimate	49.85	30.30	50.46	341.58
Std Error	1.31	5.89	7.73	2.81
 t -statistic	38.0	5.1	6.5	121.5

Why Are All t -Statistics So Strong?

In plain MLR, some predictors appeared insignificant because multicollinearity inflated their standard errors. Here, **every PC is uncorrelated with every other PC**, so each coefficient is estimated independently. Standard errors are small, t -statistics are large, and all three predictors are highly significant.

$R^2 = 0.996$ — essentially the same as the full 5-predictor model, but with only 3 predictors and no multicollinearity!

10.7 Step 7: The Verdict — MLR vs. PCR Head-to-Head

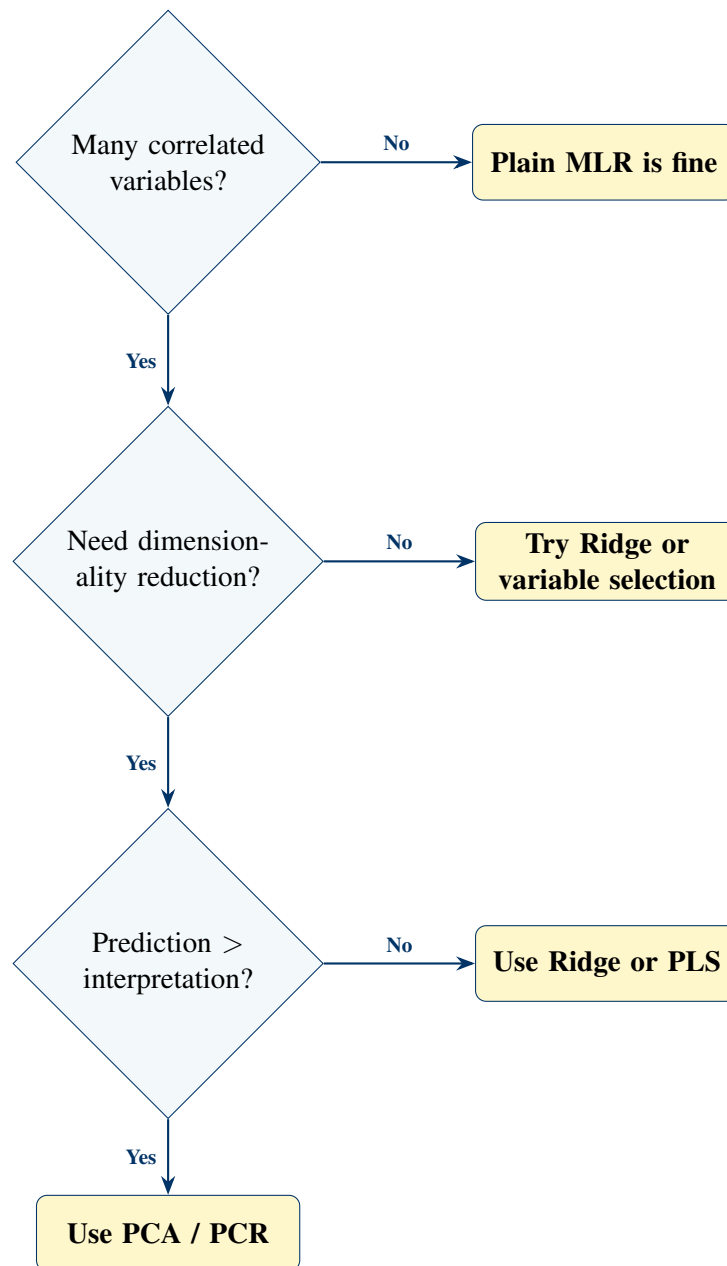
Metric	Plain MLR (5 predictors)	PCR (3 components)
R^2	0.997	0.996
Multicollinear?	YES (r up to 0.997!)	NO (PCs are orthogonal)
Coefficient stability	Unstable, inflated SE	Stable, small SE
All coefficients significant?	Some look insignificant	All $ t \gg 2$
Number of predictors	5	3 (reduced!)
Interpretability	Direct (per feature)	Abstract (use loadings)

When 5 features are nearly collinear, PCR delivers the **same predictive accuracy** ($R^2 = 0.996$ vs. 0.997) but with:

- **Fewer predictors** (3 instead of 5)
- **No multicollinearity** (orthogonal by construction)
- **Stable coefficients** (small standard errors)
- **All coefficients significant** ($|t| \gg 2$)

The trade-off: PC1 is “overall house size and newness,” not a single feature. For pure prediction, PCR is clearly superior. For explaining “how much does one bedroom add?” supplement with domain knowledge or use the loadings to translate back.

11 Decision Framework: When Should You Use PCA?



PCA Is Especially Useful When...

- You have many correlated variables and need dimensionality reduction.
- You want to visualise high-dimensional data in 2–3 dimensions.
- You want to **reduce noise** before modelling — low-variance PCs often represent noise.
- Interpretability of individual original variables is **less important** than compact representation.
- You face $p > n$ situations (more features than observations).

12 Abbreviations and Acronyms

Statistical & Mathematical Terms

Abbreviation	Full Form
PCA	Principal Component Analysis
PC	Principal Component
MLR	Multiple Linear Regression
OLS	Ordinary Least Squares
PCR	Principal Component Regression
PLS	Partial Least Squares
SVD	Singular Value Decomposition
VIF	Variance Inflation Factor
SSE	Sum of Squared Errors
RMSE	Root Mean Squared Error
CV	Cross-Validation

13 About the Author

Prof. V. Ravichandran brings over 28 years of extensive experience in corporate finance and banking, combined with 10+ years of academic excellence in financial education. As a Professor of Finance, he specialises in Financial Risk Management, Fixed Income Securities & Analysis, Value Risk and Capital Markets, and Corporate Finance.

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His educational materials are designed to meet the highest academic standards while providing practical insights drawn from real-world corporate finance and banking experience.

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